A POINT IN THE CLOSURE OF THE IMAGE OF A MORPHISM LIES IN THE SPACE OF ITS SCHEME THEORETIC IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The behaviour of the scheme theoretic image under flat base change. Invoked inside the counted proof (source lines 9001-9004). Identifies the scheme theoretic image of the base-changed morphism with the base change of the image.
- The reduction of the scheme theoretic image along a smooth cover of the source. Invoked inside the counted proof (source lines 9007-9009). Replaces the source of the base-changed morphism by a scheme.
- The statement that the image of a quasi-compact scheme onto its scheme theoretic image is dense. Invoked inside the counted proof (source lines 9010-9012) as a cross-chapter pointer. Provides density of the image of the presenting scheme.
- The description of the points of a fibre product of algebraic stacks. Invoked inside the counted proof (source lines 9005-9007) as a cross-chapter pointer. Identifies the inverse image of the space of the image with the space of the base-changed image.

Source reference. Original lemma lemma-cover-upstairs: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8754–8754. Navigation pointer only: the counted text refers to the Stacks item lemma-cover-upstairs, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-existence-plus-flat-base-change: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8885–8885. Navigation pointer only: the counted text refers to the Stacks item lemma-existence-plus-flat-base-change, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-quasi-compact-scheme-theoretic-image: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9010–9010. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-quasi-compact-scheme-theoretic-image, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-points-cartesian: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2126–2126. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-points-cartesian, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact morphism of algebraic stacks. Let $\mathcal{Z} \subset \mathcal{Y}$ be the scheme theoretic image of f. Then $|\mathcal{Z}|$ is the closure of the image of $|f|$.*

Proof. Let $z \in |\mathcal{Z}|$ be a point. Choose an affine scheme V, a point $v \in V$, and a smooth morphism $V \to \mathcal{Y}$ mapping v to z. Then $\mathcal{X} \times_{\mathcal{Y}} V$ is a quasi-compact algebraic stack. Hence we can find an affine scheme W and a surjective smooth morphism $W \to \mathcal{X} \times_{\mathcal{Y}} V$. By Lemma lemma-existence-plus-flat-base-change the scheme theoretic image of $\mathcal{X} \times_{\mathcal{Y}} V \to V$ is $Z = \mathcal{Z} \times_{\mathcal{Y}} V$. Hence the inverse image of $|\mathcal{Z}|$ in $|V|$ is $|Z|$ by Properties of Stacks, Lemma stacks-properties-lemma-points-cartesian. By Lemma lemma-cover-upstairs Z is the scheme theoretic image of $W \to V$. By Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-scheme-theoretic-image we see that the image of $|W| \to |Z|$ is dense. Hence the image of $|\mathcal{X} \times_{\mathcal{Y}} V| \to |Z|$ is dense. Observe that $v \in Z$. Since $|V| \to |\mathcal{Y}|$ is open, a topology argument tells us that z is in the closure of the image of $|f|$ as desired. □